Requirement Specification

REQ-0045 — Network Behavior — Biometric Verification

1. Identification

Requirement ID	REQ-0045
Title	Network Behavior — Biometric Verification
Type	Security
Priority	Medium
Severity	Minor
Status	Approved
Version	v2.3
Created	2022-12-13
Updated	2023-02-23
Source Document	SRS-APP-001-v3.pdf

2. Application & Solution

Application ID	APP-001
Application	SmartHome
Domain	IoT
App Version	6.0.0
Solution ID	SOL-008
Solution	Recommendation Engine
Component	Sensor Manager
Feature	data synchronization

3. Requirement Description

The system shall provide background sync consistently under stress-test conditions.

4. Acceptance Criteria

AC-1: The system shall satisfy the requirement within the defined performance SLA (p95 < 500ms).

AC-2: All failure scenarios must return a human-readable error message to the user.

AC-3: All events must be logged to the audit trail with a valid timestamp and user context.

AC-4: The feature must operate correctly in all supported environments (Mobile, Wearable, Cloud).

AC-5: Security constraints: all data in transit must be encrypted using TLS 1.3 or higher.

5. Dependencies

This requirement depends on the Recommendation Engine (SOL-008) being operational and at version 5.5 or above.

SYNTHETIC DATA — No real Samsung confidential information

6. Related Test Cases

TC-0081	Test that media streaming correctly handles invalid input wh...	Security	Critical	Draft
TC-0315	Ensure that biometric verification does not exceed the p99 l...	Sanity	Medium	Draft
TC-0385	Test that Bluetooth reconnection correctly handles invalid i...	Reliability	High	Active
TC-0489	Validate that user authentication returns the correct result...	Functional	High	Deprecated
TC-0490	Verify end-to-end flow of NFC tap-to-pay from login to pairi...	System	Critical	Active

7. Related Defects

BUG-0236	Network Failure in Sync Engine: SocketException: Connec...	Critical	Critical	REOPENED
BUG-0401	UI Failure in Payment Gateway: ElementNotFoundError: XP...	Major	High	ASSIGNED
BUG-0417	UI Failure in Bluetooth Module: ElementNotFoundError: X...	Minor	High	IN_PROGRESS

8. Synthetic Data Notice

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